Data Analytics Capstone

An AI-Assisted Decision Support Framework for Identifying and Prioritizing Healthcare Infrastructure Gaps Across Indian Districts

Interim Report

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Introduction

Healthcare infrastructure is the backbone of an effective healthcare system, offering equal access to medical facilities, healthcare professionals and essential health services. The provision of equitable healthcare infrastructure still remains a major challenge in a country like India which is geographically and demographically diverse. Rapid population growth, regional socio-economic disparities and unequal distribution of health care resources are responsible for the different levels of health care accessibility across states and districts.

In the last decade the Government of India has taken several steps to enhance the healthcare delivery by investing in Primary Health Centres (PHCs), Community Health Centres (CHCs), district hospitals and national health programs. Yet differences in healthcare infrastructure, particularly between urban and rural areas, continue to exist. With a large volume of data on healthcare, demographics and socioeconomics to analyze, decision makers are faced with the complex task of identifying the districts with the greatest need for investment in infrastructure.

The opportunities presented by data analytics and artificial intelligence (AI) provide a chance to improve evidence-based healthcare planning. In addition to descriptive reporting, explainable machine learning techniques for statistical analysis can provide predictive, data-driven decision support for policymakers. This research proposes to develop a composite Healthcare Infrastructure Gap Index (HIGI) and AI-based decision support system for identifying and ranking the districts for investment in healthcare infrastructure.

Problem Statement

India has made significant progress in expanding its public healthcare system through investments in primary, secondary and tertiary healthcare facilities. However, considerable disparities in healthcare infrastructure continue to exist across districts, resulting in an unequal access to quality healthcare services. Variations in the availability of Primary Health Centres (PHCs), Community Health Centres (CHCs), hospitals and other healthcare resources, combined with differences in population characteristics and health outcomes, make it challenging for policymakers to identify districts that require priority infrastructure investments.

Although large volumes of healthcare and demographic data are available through official government sources such as the National Family Health Survey (NFHS-5), Census of India and Rural Health Statistics (RHS), these datasets are often analysed independently using conventional statistical methods. There is limited use of integrated, data-driven approaches that combine these datasets to support objective and evidence-based infrastructure planning at the district level.

This study proposes to integrate district-level healthcare, demographic and infrastructure data to develop a **Healthcare Infrastructure Gap Index (HIGI)** that quantifies the relative infrastructure gap across districts. Artificial Intelligence (AI) and machine learning techniques will then be employed to analyse district characteristics, identify patterns among districts with similar healthcare needs, and support the prioritization of districts for future healthcare infrastructure investments. The outcome of this research is expected to provide an explainable, data-driven decision support framework that can assist policymakers in making more informed and equitable healthcare planning decisions.

Purpose of the Study

This study aims at developing and evaluating a data-driven AI assisted decision support framework for identifying and prioritizing healthcare infrastructure gaps across the districts of India. This study attempts to suggest a systematic methodological framework for the evaluation of regional inequalities in healthcare infrastructure by integrating publicly available district level healthcare, demographic and healthcare infrastructure data from official Government of India sources.

The proposed research will make a major contribution by developing a Healthcare Infrastructure Gap Index (HIGI), which is a composite index to measure and compare healthcare infrastructure gaps across districts based on multiple healthcare, demographic and infrastructure indicators. In addition, the study recommends the application of statistical analysis and unsupervised machine learning techniques to identify districts with similar healthcare infrastructure characteristics and to generate evidence-based insights for planning healthcare infrastructure and prioritizing resources.

The proposed framework intends to help policymakers and healthcare administrators in having an objective, transparent and explainable way of assessing healthcare infrastructure. The framework is meant to complement existing planning processes by allowing data-driven prioritization of districts with relatively higher infrastructure needs, rather than substitute policy decisions.

Scope of the Study

This study explores the disparity in healthcare infrastructure in the districts of India using the publicly available secondary data from the official sources of Government of India. The research limits the use of healthcare, demographic and healthcare infrastructure data sets at the district level for evidence-based assessment and prioritization of healthcare infrastructure needs.

The scope of the study includes:

- Integration of district level datasets of National Family Health Survey (NFHS-5), Census 2011 and Rural Health Statistics (RHS)
- Development of a composite Healthcare Infrastructure Gap Index (HIGI) using selected healthcare, demographic and healthcare infrastructure indicators.
- The exploration of the interrelationships of health infrastructure, demographic characteristics and health outcome indicators through statistical analysis.
- Application of unsupervised machine learning techniques including clustering methods to identify districts exhibiting similar healthcare infrastructure profiles.
- Development of an AI-supported decision support framework to support district level prioritization of health infrastructure investments
- Analytical visualizations and district-wise insights to help policymakers in health care planning and resource allocation.

The study does not include:

- Individual patient files or electronic medical records.
- Real-time healthcare surveillance or predicting future healthcare demand.
- Private healthcare infrastructure due to the limited availability of consistent district-level public data.

- Studies of economic feasibility or cost-benefit analysis of investments in health care infrastructure.
- Deployment of the proposed framework in an operational health care information system.

Aim and Objectives

Aim

The main of this capstone project is to develop an AI-assisted decision support that integrates district level healthcare, demographic, and. other healthcare infrastructure data that is publicly available from the Government of India data portal to identify, analyse, and prioritize infrastructure gaps in healthcare across the country.

Research Objectives

Research Objective 1

To develop a comprehensive analytical dataset combining the district level health care, demographic and health care infrastructure data from official sources of Government of India like National Family Health Survey (NFHS-5), Census of India 2011 and Rural Health Statistics (RHS).

Research Objective 2

To construct a composite Healthcare Infrastructure Gap Index (HIGI) to measure relative healthcare infrastructure gap across Indian districts using selected healthcare, demographic and infrastructure indicators.

Research Objective 3

To apply Artificial Intelligence (AI) and machine learning techniques to analyse the characteristics of the districts, to identify the patterns of the districts with similar healthcare infrastructure profiles and to prioritise the districts based on their relative healthcare infrastructure needs.

Research Objective 4

To develop an explainable AI-powered decision support framework that provides evidence-based information to help policymakers prioritize healthcare infrastructure investment and improve resource allocation.

Research Questions

This capstone project aims to answer the following research questions:

RQ1. Can district-level healthcare, demographic, and healthcare infrastructure data from official Government of India sources be integrated to effectively quantify healthcare infrastructure gaps across districts?

RQ2. Can a composite Healthcare Infrastructure Gap Index (HIGI) effectively measure and compare the relative healthcare infrastructure gaps among Indian districts?

RQ3. Can Artificial Intelligence and machine learning techniques identify meaningful patterns and group districts based on similarities in healthcare infrastructure, demographic characteristics, and health outcomes?

RQ4. Can an AI-assisted decision support framework improve the prioritization of districts for healthcare infrastructure investment compared to conventional data analysis approaches?

Research Hypothesis

To examine the research questions through statistical analysis, the following null and alternative hypotheses are proposed.

H01

Null Hypothesis ($H0_1$): There is no statistically significant relationship between healthcare infrastructure indicators and health outcomes across Indian districts.

Alternative Hypothesis ($H1_1$): There is a statistically significant relationship between healthcare infrastructure indicators and health outcomes across Indian districts.

H02

Null Hypothesis ($H0_2$): There is no statistically significant relationship between demographic characteristics and healthcare infrastructure availability across Indian districts.

Alternative Hypothesis ($H1_2$): There is a statistically significant relationship between demographic characteristics and healthcare infrastructure availability across Indian districts.

H03

Null Hypothesis ($H0_3$): Districts cannot be grouped into statistically distinct clusters based on healthcare, demographic, and healthcare infrastructure indicators.

Alternative Hypothesis ($H1_3$): Districts can be grouped into statistically distinct clusters based on healthcare, demographic, and healthcare infrastructure indicators.

H04

Null Hypothesis ($H0_4$): The integrated use of healthcare, demographic, and healthcare infrastructure indicators does not improve the identification and prioritization of healthcare infrastructure gaps across Indian districts.

Alternative Hypothesis ($H1_4$): The integrated use of healthcare, demographic, and healthcare infrastructure indicators improves the identification and prioritization of healthcare infrastructure gaps across Indian districts.

Data Description and Sources

Overview of Datasets

This section details the datasets employed in the study, their relevance to the research objectives and the variables chosen for analysis. By using a fusion of multiple government datasets we obtain a holistic picture of healthcare infrastructure availability, demographic characteristics and health outcomes across Indian districts.

Table 1: Overview of source Datasets

Dataset	Source	Granularity	Original Records	Records Used	Purpose
Rural Health Statistics (RHS)	Ministry of Health & Family Welfare	District	737	737	Healthcare infrastructure
Census of India 2011	Office of the Registrar General & Census Commissioner	District	640	640	Demographic characteristics
National Family Health Survey (NFHS-5)	Ministry of Health & Family Welfare / IIPS	District	707	707	Health outcome indicators

This study merges three publicly available datasets of the Government of India to develop a comprehensive analytical dataset at the district level to study the disparities in healthcare infrastructure across India.

Rural Health Statistics (RHS) provides the perspective of health care infrastructure, Census of India 2011 provides the perspective of demographic characteristics and National Family Health Survey (NFHS-5) provides the perspective of health outcome indicators.

The merging of these datasets allows a three dimensional exploration of health care access, population characteristics and health states.

Rural Health Statistics (RHS)

This subsection describes the Rural Health Statistics (RHS) dataset, its role in the study, and the healthcare infrastructure variables selected for analysis.

Table 2: Rural Health Statistics (RHS) Dataset Summary

Attribute	Description
Dataset	Rural Health Statistics (RHS)
Source	Ministry of Health and Family Welfare (MoHFW), Government of India
Granularity	District Level
Original Records	737 Districts
Records Used	737 Districts
Role in Study	Provides healthcare infrastructure indicators

Rural Health Statistics (RHS) is an official publication of Ministry of Health and Family Welfare (MoHFW), Government of India, which provides district level information on public health infrastructure in the country. The choice of the dataset is due to the presence of the main indicators of infrastructure, necessary to assess the accessibility of health care and the availability of services at the level of districts.

This study uses the RHS dataset as a main data source to quantify the healthcare infrastructure capacity and is part of the infrastructure component of the proposed Healthcare Infrastructure Gap Index (HIGI).

Table 3: Selected RHS Variables

Variable	Description	Purpose
Sub Centres	Number of Sub Centres	Primary healthcare infrastructure
PHCs	Number of Primary Health Centres	Primary healthcare delivery
CHCs	Number of Community Health Centres	Secondary healthcare delivery
Sub-Divisional Hospitals	Number of Sub-Divisional Hospitals	Higher-level healthcare facilities
District Hospitals	Number of District Hospitals	District-level referral hospitals

Only infrastructure-related variables that were directly involved in the assessment of the availability of healthcare facilities were retained. During preprocessing, administrative fields and variables not relevant to the research objectives were deleted.

The selected variables together depict the hierarchical structure of the public healthcare delivery system in India and also provide the infrastructure indicators required for further exploratory analysis and construction of the Healthcare Infrastructure Gap Index (HIGI).

Census of India 2011

This subsection describes the Census of India 2011 dataset, its contribution to the study, the preprocessing performed to obtain district-level records, and the demographic variables selected for analysis.

Table 4: Census of India 2011 Dataset Summary

Attribute	Description
Dataset	Census of India 2011
Source	Office of the Registrar General & Census Commissioner, India
Granularity	District Level
Original District Records	640 Districts
Records Used	640 Districts
Role in Study	Provides demographic indicators representing healthcare demand

Official national census is the Census of India 2011, conducted by the office of the Registrar General and Census Commissioner, India. It contains a wealth of demographic, social and economic information at several levels of administration.

For this study, the dataset was preprocessed to include only district level records representing total population . Specifically, the records with Level = DISTRICT and TRU = Total were selected to maintain consistency with the district-level Rural Health Statistics (RHS) and National Family Health Survey (NFHS-5) datasets. This preprocessing resulted in a standardized demographic dataset ready for integration.

Table 5: Selected Census Variables

Variable	Description	Purpose
TOT_P	Total Population	Represents population size and healthcare demand
P_LIT	Literate Population	Indicates educational attainment within the district
P_ILL	Illiterate Population	Represents the population requiring additional consideration in healthcare accessibility
No_HH	Number of Households	Represents household distribution and population density

The selected demographic variables are proxies of the potential district-level demand for healthcare services. The size of the population requiring health care services is measured by population and household counts, and demographic context that may affect health care awareness, service utilisation and health outcomes is provided by literacy and illiteracy levels.

These variables complement the infrastructure indicators extracted from the RHS dataset and allow for further exploratory analysis and modelling.

National Family Health Survey (NFHS-5)

This subsection describes the National Family Health Survey (NFHS-5) dataset, the health indicators selected for the study, and their relevance to assessing healthcare accessibility and population health outcomes.

Table 6: National Family Health Survey (NFHS-5) Dataset Summary

Attribute	Description
Dataset	National Family Health Survey (NFHS-5)
Source	International Institute for Population Sciences (IIPS) and Ministry of Health & Family Welfare (MoHFW), Government of India
Survey Period	2019–2021
Granularity	District Level
Original Records	707 Districts
Records Used	707 Districts
Role in Study	Provides district-level healthcare utilisation and health outcome indicators

The National Family Health Survey (NFHS-5) is a nationally representative health survey conducted by the International Institute for Population Sciences (IIPS) under the stewardship of the Ministry of Health and Family Welfare (MoHFW), Government of India. The survey provides detailed district-level data on maternal and child health, nutrition, health care utilization, disease prevalence and health related behaviors.

A subset of health outcome indicators was selected to evaluate access to healthcare, service use and population health for this study. These variables complement the healthcare infrastructure indicators from the RHS dataset and the demographic indicators from the Census dataset to allow for a multidimensional assessment of healthcare disparities across the districts.

Table 7: Selected NFHS-5 Variables

Variable	Description	Purpose
Health_Insurance	Population covered under a health insurance or financing scheme (%)	Represents financial access to healthcare
Institutional_Births	Births occurring in healthcare institutions (%)	Measures utilisation of institutional maternal healthcare services
Skilled_Birth_Attendance	Births attended by skilled healthcare personnel (%)	Indicates access to skilled maternal healthcare
Full_Immunization	Fully immunized children (%)	Measures preventive child healthcare coverage
Stunting	Children under five years who are stunted (%)	Indicates chronic child malnutrition
Underweight	Children under five years who are underweight (%)	Represents nutritional status of children
Women_Anaemia	Women aged 15–49 years who are anaemic (%)	Indicates women's nutritional and public health status

The selected NFHS-5 variables represent key indicators of healthcare accessibility, service utilisation, and health outcomes. Together, they provide insights into maternal and child health, preventive healthcare coverage, nutritional status, and financial access to healthcare services.

These indicators complement the infrastructure and demographic variables obtained from the RHS and Census datasets, enabling a comprehensive assessment of healthcare disparities across districts.

Integrated Analytical Dataset

The three source datasets were merged to create a harmonized analytical dataset at the district level, suitable for exploring disparities in healthcare infrastructure across India. The datasets came from different governmental sources and had inconsistencies in the naming conventions of the districts and their administrative classifications. Hence, a systematic preprocessing and harmonization process was accomplished before integration.

Data cleaning and harmonization processes were used to standardize district names across all of the datasets. The Census dataset was filtered to retain only district level records representing total population and relevant healthcare infrastructure and health outcome variables were selected from RHS and NFHS-5 datasets respectively. The datasets were preprocessed and merged on the common identifiers of State and District.

The integration process resulted in a consolidated analytical dataset of 488 common districts with 18 variables on healthcare infrastructure, demographic characteristics, and health outcome indicators. All the subsequent exploratory analysis, visualization and the proposed Healthcare Infrastructure Gap Index (HIGI) framework are based on this integrated dataset.

Table 8: Categories of Variables in the Integrated Dataset

Category	Variables	Number of Variables
Healthcare Infrastructure	Sub Centres, PHCs, CHCs, Sub-Divisional Hospitals, District Hospitals	5
Demographic Characteristics	Total Population, Literate Population, Illiterate Population, Number of Households	4
Health Outcome Indicators	Health Insurance, Institutional Births, Skilled Birth Attendance, Full Immunization, Stunting, Underweight, Women Anaemia	7
Identifiers	State, District	2

Table 9: Integration Summary

Item	Value
RHS Records	737
Census Records	640
NFHS Records	707
Final Integrated Dataset	**488 Districts × 18 Variables**

The integrated analytical dataset combines complementary infrastructure, demographic, and health outcome indicators into a single district-level dataset, providing a robust and comprehensive foundation for exploratory analysis and the proposed modelling framework.

Data Dictionary

This subsection defines the variables retained in the integrated analytical dataset, providing their descriptions, data types, source datasets, and analytical purpose. The data dictionary enhances transparency and reproducibility by documenting the variables used throughout the study.

Table 10: Data Dictionary of the Integrated Analytical Dataset

Variable	Description	Data Type	Source	Analytical Category
State	State or Union Territory	Categorical	Common Identifier	Identifier
District	District Name	Categorical	Common Identifier	Identifier
Sub_Centres	Number of Sub Centres	Numeric	RHS	Healthcare Infrastructure
PHCs	Number of Primary Health Centres	Numeric	RHS	Healthcare Infrastructure
CHCs	Number of Community Health Centres	Numeric	RHS	Healthcare Infrastructure
Sub_Divisional_Hospital	Number of Sub-Divisional Hospitals	Numeric	RHS	Healthcare Infrastructure
District_Hospital	Number of District Hospitals	Numeric	RHS	Healthcare Infrastructure
TOT_P	Total Population	Numeric	Census	Demographic
P_LIT	Literate Population	Numeric	Census	Demographic
P_ILL	Illiterate Population	Numeric	Census	Demographic
No_HH	Number of Households	Numeric	Census	Demographic
Health_Insurance	Population covered by health insurance (%)	Numeric	NFHS-5	Health Outcome

Institutional_Births	Institutional births (%)	Numeric	NFHS-5	Health Outcome
Skilled_Birth_Attendance	Births attended by skilled health personnel (%)	Numeric	NFHS-5	Health Outcome
Full_Immunization	Fully immunized children (%)	Numeric	NFHS-5	Health Outcome
Stunting	Children under five years who are stunted (%)	Numeric	NFHS-5	Health Outcome
Underweight	Children under five years who are underweight (%)	Numeric	NFHS-5	Health Outcome
Women_Anaemia	Women aged 15–49 years who are anaemic (%)	Numeric	NFHS-5	Health Outcome

The final integrated dataset consists of **18 variables**, including **2 identifier variables**, **5 healthcare infrastructure variables**, **4 demographic variables**, and **7 health outcome variables**.

These variables collectively represent the multidimensional characteristics required to evaluate healthcare infrastructure disparities and support the exploratory analysis and proposed modelling framework presented in this study.

Project Repository

To promote transparency and reproducibility, all project artefacts developed during this study are maintained in a GitHub repository. The repository serves as the central location for the analytical workflow, project documentation, and version-controlled source code used throughout the study.

The repository includes:

- Jupyter Notebook containing data preprocessing, integration, exploratory data analysis (EDA), and visualizations.
- Interim report and supporting documentation.

- Dataset references and project documentation.
- Source code developed during the analytical workflow.

The project repository can be accessed using the following link:

GitHub Repository: https://github.com/Haraani/Healthcare-Infrastructure-Gap-Index

Research Methodology

The study uses a quantitative, data driven research methodology based on secondary datasets available in public domain from official sources of Government of India. The study uses data preparation, exploratory data analysis (EDA), statistical analysis and machine learning techniques to examine inequities in healthcare infrastructure across Indian districts.

The methodology consists of seven continuous stages which are research design, data preparation, exploratory data analysis, Healthcare Infrastructure Gap Index (HIGI) construction, statistical analysis, machine learning analysis, evaluation and validation, and the development of an AI-assisted decision support framework.

The overall methodology comprises seven consecutive stages:

1. Research design
2. Data preparation
3. Healthcare Infrastructure Gap Index (HIGI) construction
4. Statistical analysis
5. Machine learning analysis
6. Evaluation and validation and
7. Development of the decision support framework.

Research Design

This study has adopted a quantitative observational research design with secondary data collected from publicly available sources of the Government of India. Districts are the unit of analysis, enabling a comparative assessment of the availability of healthcare infrastructure, demographic characteristics, and health outcomes across India.

The study combines data from Rural Health Statistics (RHS), Census of India 2011 and National Family Health Survey (NFHS-5) to build a complete analytical data set at the district level. The methodology combines data preparation, exploratory data analysis, statistical techniques and machine learning methods to identify disparities in infrastructure and to help develop the proposed Healthcare Infrastructure Gap Index (HIGI).

Data Preparation and Integration

Prior to analysis, the three source datasets were systematically preprocessed to ensure their consistency, completeness, and compatibility for district-level integration. Preprocessing activities comprised filtering the datasets, selecting the variables, harmonizing the district names, validating the data types and assessing the missing values.

Only district level records for total population from the dataset of Census of India was kept. Healthcare Infrastructure Variables Relevant infrastructure variables were extracted from the Rural Health Statistics (RHS) dataset. Health Outcome Variables Selected health outcome indicators were retained from the National Family Health Survey (NFHS-5).

District Names District names were standardized because there were variations in nomenclature and administrative representations, since data were collected from a variety of government sources. The datasets were harmonized and merged on the basis of common identifiers of State and District to create an analytical dataset with 488 common districts and 18 variables.

Prior to exploratory analysis, data quality procedures were applied to the integrated dataset; duplicates, missing values, inconsistent data types and invalid data entries were identified and removed.

Table 11: Data Cleaning Summary

Activity	Outcome
Dataset filtering	District-level records retained
Variable selection	Relevant infrastructure, demographic and health variables selected
District harmonization	District names standardized across datasets
Dataset integration	Performed using State and District
Duplicate check	No duplicate records in final dataset
Missing value assessment	Completed before analysis
Final analytical dataset	488 districts × 18 variables

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the integrated dataset prior to statistical modelling. The objective of this stage was to evaluate data quality, examine variable distributions, identify missing values and outliers, and explore potential relationships among the selected variables.

The EDA included descriptive statistical analysis, missing value assessment, distribution analysis, and correlation analysis. These analyses provided an initial understanding of the healthcare infrastructure, demographic characteristics, and health outcome indicators across the 488 districts included in the final analytical dataset.

The findings from the EDA informed subsequent feature selection, model development, and the proposed Healthcare Infrastructure Gap Index (HIGI) framework.

Construction of Healthcare Infrastructure Gap Index (HIGI)

A composite index, the Healthcare Infrastructure Gap Index (HIGI), is proposed to measure disparities in healthcare infrastructure across districts of India. Individual indicators are useful in providing insight into specific aspects of healthcare delivery but they do not capture the multidimensional nature of healthcare infrastructure . HIGI addresses this limitation by integrating healthcare infrastructure, demographic characteristics, and health outcome indicators into a single measure at the district level, allowing for comparisons across regions.

The proposed index is based on three key dimensions which are an important component in the assessment of healthcare infrastructure.

Table 12: Dimensions of HIGI

Dimension	Variables Included
Healthcare Infrastructure	Sub Centres, Primary Health Centres (PHCs), Community Health Centres (CHCs), Sub-Divisional Hospitals, District Hospitals
Demographic Characteristics	Total Population, Literate Population, Illiterate Population, Number of Households
Health Outcome Indicators	Health Insurance Coverage, Institutional Births, Skilled Birth Attendance, Full Immunization, Stunting, Underweight, Women Anaemia

These measures provide a broad view of health care access, population characteristics, and health outcomes. Rather than relying on a single indicator, the HIGI combines indicators from different sectors to provide a comprehensive assessment of gaps in healthcare infrastructure.

The construction of the HIGI will follow a step-by-step methodology, including data pre-processing, normalization of features and integration of selected indicators into a composite index. The resulting HIGI scores will enable district comparisons, identify underserved areas and support evidence-based healthcare planning and resource allocation.

The Chapter Modelling describes in detail the methodology of feature selection, weighting, index construction and validation.

Statistical Analysis

The objective of the statistical analysis stage is to understand the relationships between health infrastructure, demographic characteristics, and health outcomes among districts in India. The analyses will provide quantitative evidence to answer the research questions and to help in the construction and validation of the proposed Healthcare Infrastructure Gap Index (HIGI).

The statistical analysis will be performed in two steps. The first phase includes descriptive statistical analysis and exploratory data analysis (EDA) that have been completed to provide an overview of the characteristics of the integrated dataset, assess data quality and understand the distribution of the selected variables. The second phase will use inferential statistical methods, such as correlation analysis, hypothesis testing, and regression analysis, to examine relationships between variables and test the significance of observed relationships.

The results of the statistical analysis will help in feature selection for the HIGI and in the development of predictive models and will provide quantitative evidence for healthcare planning and decision-making at the district level.

AI and Machine Learning Methodology

Machine learning approaches will be applied to detect patterns in disparities in healthcare infrastructure and to aid data-driven decision making. The Healthcare Infrastructure Gap Index(HIGI) will be constructed and standard ML models will be applied to identify similarities across districts and classify them based on infrastructure characteristics and identify the factors associated with healthcare access and health outcomes.

The modelling approach will be determined by the integrated dataset and the nature of the research questions. Proper feature selection methods will be employed to choose the most informative variables. Data preparation techniques such as normalization and feature scaling will be implemented, if required, to enhance a model's performance.

We will discuss both supervised and unsupervised learning approaches. Clustering (an unsupervised learning approach) will be used to cluster the districts with similar healthcare infrastructure profiles and identify regions with similar healthcare needs. Then, models with supervised learning can be used to predict categories of healthcare infrastructure or specific health outcome indicators from infrastructure and demographic characteristics.

The performance of the model will be evaluated by using appropriate validation techniques and performance measures for the selected algorithms. The machine learning analysis will enable statistical analysis to detect complex relationships in the data and provide further evidence to support healthcare planning and resource allocation.

Table 13: Proposed ML Models

Model	Purpose	Justification
K-Means Clustering	Group districts with similar healthcare profiles	Appropriate for identifying natural clusters without predefined labels
Hierarchical Clustering	Validate cluster structure	Provides interpretable cluster relationships
Random Forest	Feature importance and prediction	Handles non-linear relationships and provides feature importance
Linear Regression	Predict health outcome indicators	Simple and interpretable baseline model

Evaluation and Validation

We will evaluate the reliability, interpretability and usefulness of the proposed analytical framework to support district-level healthcare planning. Appropriate statistical and machine learning metrics for performance will be used to evaluate the performance of the chosen modelling techniques. The goal is to ensure that the models developed provide meaningful insights, while being robust and generalizable.

For supervised learning models, performance will be measured by metrics (for example, accuracy, precision, recall, F1-score, and AUC-ROC (Area Under the ROC Curve) where applicable. For regression-based models, evaluation will be based on metrics including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and coefficient of determination (R^2). For the unsupervised learning models we will use cluster validation measures such as the Silhouette Score and Davies-Bouldin Index to evaluate the quality and separation of clusters.

The proposed Healthcare Infrastructure Gap Index (HIGI) will be tested for consistency and interpretability using sensitivity analysis and comparison with known healthcare indicators. The evaluation process will ensure that the proposed framework provides reliable evidence that can be used to assess healthcare infrastructure, and to inform policy decisions.

Mapping of Research Questions to Analytical Approach

The four research questions defined in this study cover different areas of healthcare infrastructure assessment, from the understanding of infrastructure disparity to the support of evidence-based decision making. Each research question is linked to a suitable analytical methodology, to ensure the appropriateness of the techniques selected for the study objectives.

Table 14 shows the relationship between the research questions, the proposed analytical approaches and the expected outcomes.

Table 14: Mapping of Research Questions to Analytical Approach

	Analytical Approach	Expected Outcome
RQ1	Exploratory Data Analysis (EDA), Descriptive Statistics, Correlation Analysis and Regression Analysis	Identification of statistically significant relationships between healthcare infrastructure and health outcomes.
RQ2	Feature Selection, Data Normalization, Composite Index Construction and Validation	Development of a standardized district-level Healthcare Infrastructure Gap Index (HIGI).
RQ3	Unsupervised Learning (Clustering) and Supervised Machine Learning Models	Identification of district clusters and predictive models supporting infrastructure assessment.
RQ4	Integration of HIGI, Statistical Findings and Machine Learning Outputs into an AI-Assisted Decision Support Framework	A data-driven framework supporting healthcare planning, resource prioritization and policy decision-making.

The analytical approaches outlined above provide a structured framework for addressing each research question while ensuring alignment with the study objectives. The combination of exploratory analysis, statistical methods, composite index development, and machine learning techniques enables a comprehensive assessment of healthcare infrastructure disparities and supports the development of evidence-based decision-support tools for healthcare planning.

Analysis

Data Cleaning

The integrated analytical dataset was subjected to a systematic data cleaning process to ensure data quality and consistency before the exploratory analysis and modelling. The goal of this step was to find inconsistencies, confirm data types, look for missing values and ensure the integrity of the merged dataset.

The data integration preprocessing activities included district name harmonization, variable selection, data type validation, and duplicate record assessment and missing value analysis. These steps made sure that the final dataset was fit for statistical analysis and later modelling.

The final analytical dataset contained 488 districts and 18 variables on health infrastructure, demographic characteristics and health outcome indicators.

Table 15: Data Cleaning Summary

Data Cleaning Activity	Outcome
District name harmonization	Completed
Variable selection	Completed
Dataset integration	Completed
Duplicate record assessment	No duplicate records identified
Data type validation	Completed
Missing value assessment	Completed
Final analytical dataset	488 districts × 18 variables

Missing Value Assessment

The integrated dataset was examined for missing observations across all variables. The assessment indicated that most variables contained complete information. Missing values were identified only in three health outcome variables.

Table 16: Missing Value Assessment

Variable	Missing Values
Full Immunization	9
Stunting	1
Underweight	1

No missing values were observed in the healthcare infrastructure or demographic variables. Given the very small proportion of missing observations (less than 2% of the dataset), the missing values were retained for subsequent analysis without significantly affecting the overall integrity of the dataset.

Overall, the data cleaning process confirmed that the integrated dataset was of high quality and suitable for exploratory analysis and modelling.

The limited number of missing observations and the absence of duplicate records indicate that the dataset provides a reliable foundation for subsequent statistical analysis and machine learning.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the integrated analytical dataset and identify underlying patterns, relationships, and potential anomalies before statistical modelling. The analysis focused on evaluating data quality, summarizing the distribution of variables, examining relationships among healthcare infrastructure and health outcome indicators, and generating preliminary insights to support subsequent modelling.

The EDA comprised descriptive statistical analysis, missing value assessment, distribution analysis, and correlation analysis. The findings provide an overview of the district-level healthcare infrastructure landscape and establish the analytical foundation for the proposed Healthcare Infrastructure Gap Index (HIGI).

Descriptive Statistics

Descriptive statistics were calculated to present the central tendency, spread and range of each numeric variable included in the study. These statistics give a general idea of the variation that exists across the 488 districts.

The descriptive analysis indicates large differences across districts in availability of health care infrastructure, population characteristics and health outcomes. The large ranges of several variables indicate considerable regional variation and therefore warrant the need for a composite infrastructure index that can capture such variations.

Distribution Analysis

Distribution analysis was done to analyze the frequency distribution, variability and skewness of the selected variables. To explore the distributional properties of the data further, histograms were separately constructed for healthcare infrastructure and demographic and health outcome variables.

Healthcare Infrastructure Variables

The variables of health care infrastructure are positively skewed with the majority of the districts having relatively lower number of health care facilities while a few districts have a large number of health care infrastructure. This pattern suggests that there is a lot of disparity in the availability of health care infrastructure across districts.

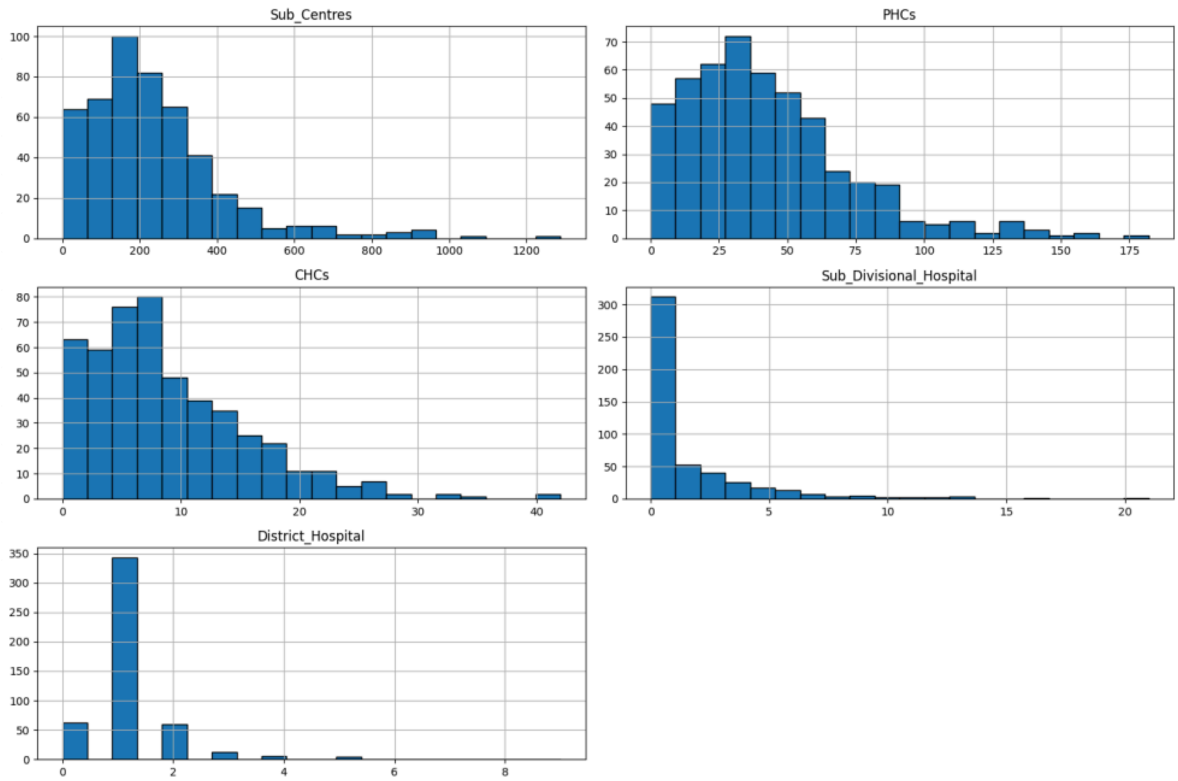

Figure 1: Distribution of Healthcare Infrastructure Variables

Demographic Variables

The demographic indicators are also characterized by different degrees of dispersion. Population variables show strong right skewness due to the presence of districts with high population density. Literacy related indicators show comparatively less variation. Such differences reflect the demographic heterogeneity across districts in India.

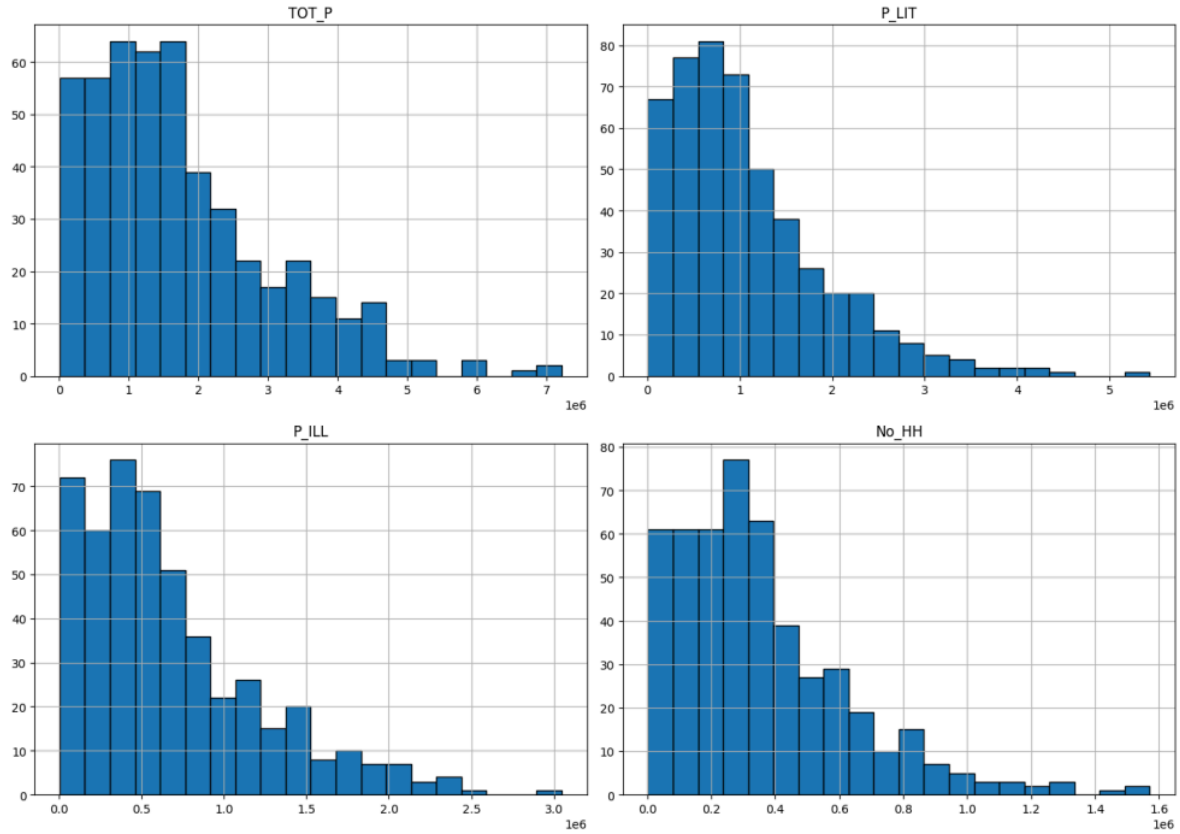

Figure 2: Distribution of Demographic Variables

Health Outcome Variables

In general, health outcome indicators have more concentrated distributions than infrastructure variables. Many districts have relatively high values for institutional births, skilled birth attendance and full immunization, while the wide variation observed in indicators such as stunting, underweight prevalence and women anemia suggests the persistence of disparities in maternal and child health outcomes.

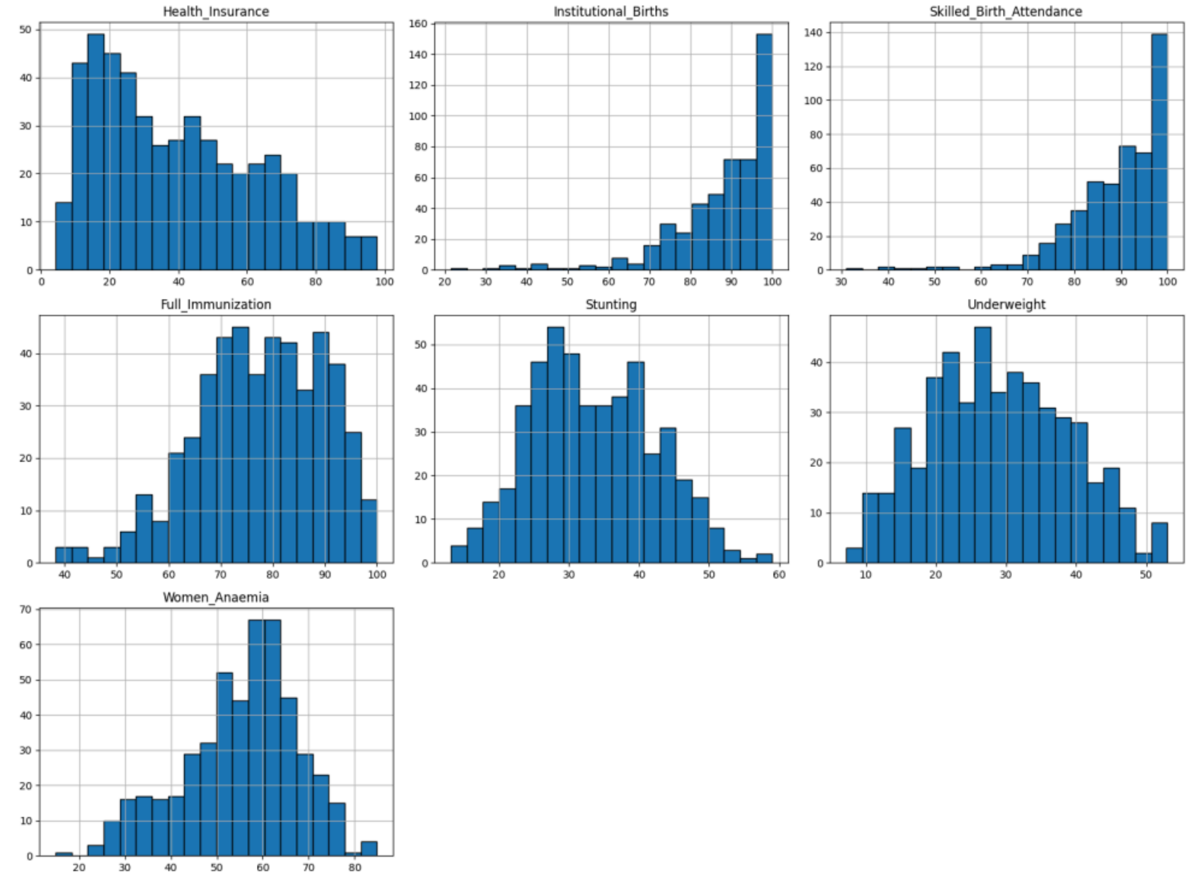

Figure 3: Distribution of Health Outcome Variables

Correlation Analysis

A Pearson correlation analysis was conducted to explore the relationships between healthcare infrastructure, demographic characteristics and selected health outcome indicators. The correlation matrix obtained gives an overall picture of the strength and direction of pairwise relationships between the variables included in the study.

The analysis shows a strong positive correlation among the variables of healthcare infrastructure such as Sub Centres, Primary Health Centres (PHCs), Community Health Centres (CHCs) and district population signifying that the districts with higher population generally have higher healthcare infrastructure capacity.

Healthcare infrastructure was also found to have a positive association with favorable health outcome indicators like Institutional Births and Skilled Birth Attendance. In contrast, Stunting and Underweight as negative health indicators have negative relationships with some of the healthcare infrastructure variables, meaning that better availability of healthcare may be associated with better maternal and child health outcomes.

Correlation analysis provides preliminary evidence supporting the study hypothesis that availability of healthcare infrastructure influences district-level health outcomes.

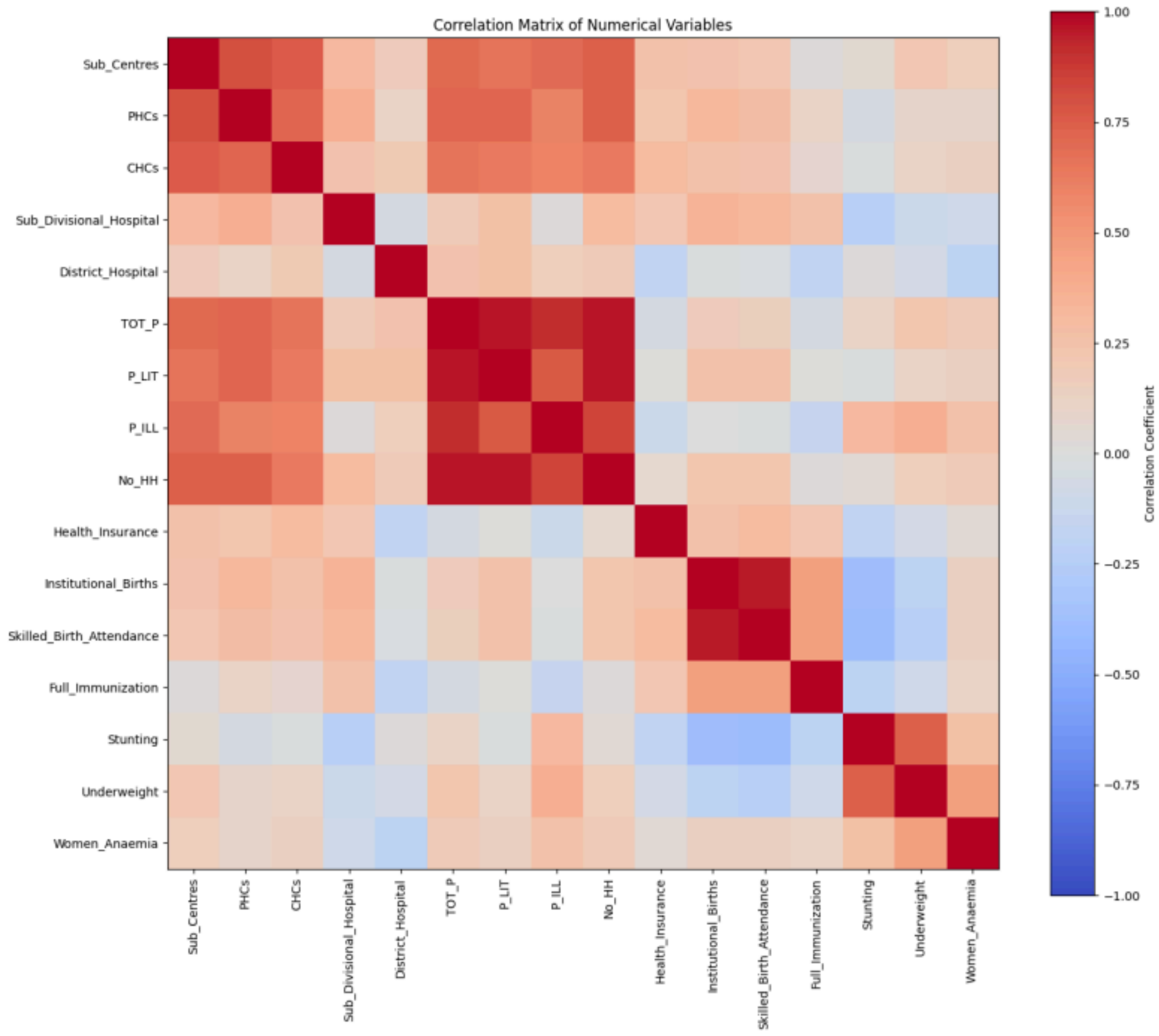

Figure 4: Correlation Matrix of the Integrated Analytical Dataset

Key Insights

The exploratory analysis produced several important findings that provide direction for the subsequent modelling phase:

- Considerable disparities exist in healthcare infrastructure availability across Indian districts.
- Population size is positively associated with the availability of healthcare facilities.
- Districts with better healthcare infrastructure generally exhibit improved maternal and child health indicators.
- Child malnutrition indicators such as Stunting and Underweight demonstrate negative associations with several healthcare infrastructure variables.
- The integrated dataset exhibits high data quality with minimal missing observations, making it suitable for statistical analysis and machine learning.
- The observed relationships provide a strong basis for constructing the Healthcare Infrastructure Gap Index (HIGI) and developing predictive models in the subsequent chapter.

Proposed Modelling Framework

Feature Selection

Feature selection is a critical step in developing robust analytical and machine learning models, as it ensures that only variables relevant to the research objectives are retained for analysis. The proposed feature set is derived from the integrated dataset based on the relevance of each variable to healthcare infrastructure availability, demographic characteristics, and health outcomes.

The study incorporates three categories of features that collectively provide a comprehensive representation of district-level healthcare systems. Healthcare infrastructure variables quantify the availability of healthcare facilities, demographic variables capture population characteristics and service demand, while health outcome indicators reflect the effectiveness and accessibility of healthcare services.

Construction of the Healthcare Infrastructure Gap Index (HIGI)

This study proposes the Healthcare Infrastructure Gap Index (HIGI) as a composite indicator to quantify the gaps in healthcare infrastructure across districts of India. Instead of focusing on individual indicators, the index aggregates various aspects of healthcare infrastructure, demographic factors, and health outcomes into a single standardized metric.

Construction of the HIGI will be performed through data pre-processing, feature normalization and aggregation of standardized indicators to a composite score. The resulting HIGI score will enable district level comparison, identification of underserved areas and support evidence based healthcare planning.

The proposed HIGI is intended to provide policymakers with an interpretable measure of adequacy of healthcare infrastructure that can be useful in identifying priority districts for infrastructure investment and resource allocation.

Machine Learning Model Selection

The statistical analysis will be supplemented by machine learning methods to find patterns in the integrated healthcare dataset and to help in data-driven decision-making. The choice of model depends on the nature of research questions, and the characteristics of data at hand.

Table 17: Proposed Machine Learning Models and Their Justification

Model	Purpose	Justification
K-Means Clustering	Group districts with similar healthcare profiles	Suitable for discovering natural clusters without predefined class labels
Supervised Learning Models (to be finalized)	Predict healthcare infrastructure gaps or health outcomes	The final supervised model will be selected based on data characteristics and project requirements following exploratory analysis.

The proposed modelling framework combines unsupervised and supervised learning techniques. Clustering methods are proposed to identify districts with similar healthcare characteristics, while appropriate supervised learning models will be finalized following feature engineering and exploratory analysis.

Model Evaluation Metrics

The performance of the proposed models will be assessed using evaluation metrics appropriate to the modelling technique employed. Different metrics will be used depending on whether the task involves clustering, regression, or classification.

Table 18: Proposed Model Evaluation Metrics

Model Type	Evaluation Metrics
Clustering	Silhouette Score, Davies–Bouldin Index
Regression	Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Coefficient of Determination (R^2)
Classification *(if implemented)*	Accuracy, Precision, Recall, F1-Score, AUC-ROC

Preliminary Results

The interim phase of the study has successfully completed data acquisition, preprocessing, dataset integration, and exploratory data analysis. Three publicly available Government of India datasets i.e., Rural Health Statistics (RHS), Census of India 2011, and National Family Health Survey (NFHS-5) were integrated to create a consolidated district-level analytical dataset comprising **488 districts** and **18 variables**.

The completed analyses establish a strong foundation for the subsequent development of the Healthcare Infrastructure Gap Index (HIGI) and the proposed machine learning models.

Preliminary Findings

The exploratory analysis has produced several initial observations:

- The integrated dataset demonstrates high data quality with only a minimal number of missing observations.

- Considerable variation exists in healthcare infrastructure availability across Indian districts.
- Demographic characteristics vary substantially across districts, indicating differences in healthcare demand.
- Health outcome indicators exhibit regional variation, suggesting disparities in healthcare accessibility and service delivery.
- Initial correlation analysis indicates meaningful relationships between healthcare infrastructure availability and selected maternal and child health indicators.

These findings support the feasibility of constructing a composite Healthcare Infrastructure Gap Index (HIGI) for district-level comparison and prioritization.

Work Planned for the Final Phase

The remaining phase of the study will focus on:

- Construction and validation of the Healthcare Infrastructure Gap Index (HIGI).
- Implementation of the proposed machine learning models.
- Model evaluation and comparison using appropriate performance metrics.
- Interpretation of modelling results.
- Development of recommendations to support evidence-based healthcare planning and policy decision-making.

The completion of these activities will enable a comprehensive assessment of healthcare infrastructure disparities and fulfil the objectives outlined in this study.

Recommendation and Application

The proposed AI assisted decision support framework intends to support evidence-based healthcare infrastructure planning, and provide systematic approaches to policymakers and healthcare administrators to identify districts requiring priority healthcare infrastructure investments.

The objective of this study is to enhance the transparency and consistency of district-level healthcare infrastructure assessment by combining healthcare, demographic and healthcare infrastructure indicators into a single analytical framework.

The proposed Healthcare Infrastructure Gap Index (HIGI) together with statistical analysis and machine learning based district segmentation can help decision makers to compare healthcare infrastructure gaps across districts and to identify districts with similar healthcare characteristics.

These analytical insights could help inform planning, help to prioritize investments in healthcare infrastructure, and help to allocate healthcare resources equitably.

The main target users of the proposed framework are:

- Government of India Ministry of Health and Family Welfare (MoHFW)
- National Health Mission (NHM)
- State Dept. of Health
- District Health Authorities
- Policy analysts and public health planners
- Researchers in healthcare infrastructure planning and public health analytics.

The proposed framework aims to complement existing planning processes by providing a transparent, explainable, and data-driven approach to healthcare infrastructure assessment. Although the framework is developed using the Indian district level data, the proposed methodology can also be adapted for similar healthcare infrastructure assessment studies in other geographical regions where similar public datasets are available.

Significance of the study

Healthcare infrastructure is of critical importance in ensuring equal access to quality healthcare services. Identification of the areas with poor healthcare infrastructure is important for effective healthcare planning and efficient utilization of limited public resources.

This research contributes to the domain of healthcare analytics by integrating healthcare, demographic and infrastructure data from various official sources of Government of India to present a comprehensive assessment of healthcare infrastructure gaps at the district level.

The proposed AI-assisted decision support framework is expected to help policymakers and healthcare administrators with evidence-based prioritization of districts for healthcare infrastructure investments. Further, the Healthcare Infrastructure Gap Index (HIGI) being proposed as part of this study can provide a systematic way to compare healthcare infrastructure disparities across districts.

From an academic point of view, the study illustrates the application of data analytics and Artificial Intelligence in solving real-world public health problems using government datasets. The proposed methodology can also be applied to similar infrastructure planning and resource allocation problems in other public sectors.

Limitations of the Study

The study has the following limitations:

- The analysis is based entirely on data obtained from official Government of India sources. Consequently, the quality and completeness of the results depend on the accuracy of the published datasets.
- The datasets used in this study originate from different years, including the National Family Health Survey (2019–21), Census of India 2011, and Rural Health Statistics.

Temporal differences between datasets may influence the analysis and interpretation of results.

- The study is limited to publicly available district-level healthcare, demographic, and infrastructure indicators and does not incorporate real-time healthcare data or private healthcare infrastructure.
- The proposed AI-assisted framework is intended to support evidence-based decision-making and should be considered a decision support tool rather than a replacement for policy decisions or expert judgment.
- The findings of the study are specific to the Indian healthcare system and the selected datasets and may not be directly generalizable to other countries or healthcare systems.

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